EchoChain – Power BI Analytics & Visualization

1. What is Power BI?

Power BI is a Microsoft tool used to analyze data and create interactive reports and dashboards.

In EchoChain, Power BI is used to present the project data in an easy way so that a business user can understand product lifecycle, warranty and secondary-market information.

The basic flow I learned is:

Data → Power Query → Data Model → DAX → Visualizations → Dashboard → Business Insights

---

2. Power BI Basics

I learned the following basic Power BI concepts:

- Power BI Desktop
- Power BI Service
- Reports
- Dashboards
- Data sources
- Visuals
- Filters
- Slicers
- Pages

Power BI Desktop is mainly used to prepare data, create relationships, write calculations and build reports.

Power BI Service is used to publish and share reports online.

---

3. Connecting Power BI with Databricks

For EchoChain, Power BI needs to connect with the data available through Databricks.

The basic connection is:

Databricks → SQL Warehouse / SQL Endpoint → Power BI

I learned about two ways of working with data:

Import

Data is loaded into Power BI.

DirectQuery

Power BI sends queries to the original data source when the report is being used.

For my project, I mainly focused on understanding and setting up the connection between **Databricks SQL and Power BI**.

---

4. Power Query

Power Query is used for cleaning and preparing data before creating the report.

I learned about:

- Removing unnecessary columns
- Renaming columns
- Changing data types
- Removing duplicates
- Handling missing values
- Filtering rows
- Replacing values
- Splitting columns
- Merging tables
- Appending tables

For example, if a price column is treated as text, I can change it to a number so that it can be used for calculations.

I also learned that **Merge** is similar to joining tables, while **Append** is used to combine rows from different tables.

---

5. Data Modeling

Data modeling is important because Power BI needs to know how different tables are connected.

For EchoChain, the project can have information related to:

- Products
- Components
- BOM
- Warranty
- Secondary-market listings

These tables can be connected using common fields such as SKU or product information.

Important terms

Primary Key: A column that uniquely identifies a record.

Foreign Key: A column that connects to a key in another table.

Relationship: The connection between two tables.

I also learned about:

- One-to-one relationship
- One-to-many relationship
- Many-to-many relationship

One-to-many is one of the most commonly used relationships.

---

6. Star Schema

I learned that a star schema is a common way of organizing data for reporting.

It mainly contains:

Dimension Tables

These describe things such as:

- Product
- Component
- Date

Fact Tables

These contain records or events such as:

- Warranty information

- BOM information
- Secondary-market listings

A simple EchoChain model can be thought of as:

Product / Component / Date → Warranty / BOM / Secondary Market

The exact tables depend on the final project data provided by the team.

---

7. DAX

DAX stands for **Data Analysis Expressions**.

It is used in Power BI to create calculations and measures.

The basic DAX functions I learned are:

- SUM
- COUNT
- COUNTROWS
- DISTINCTCOUNT
- AVERAGE
- DIVIDE
- IF
- FILTER
- CALCULATE

For example:

```
Total Listings = COUNTROWS(EBAY)


Average Resale Price = AVERAGE(EBAY[Price])


Unique Products = DISTINCTCOUNT(Products[SKU])
```

I learned that **CALCULATE** is an important DAX function because it can change the filter used for a calculation.

---

8. Measures and Calculated Columns

I learned the difference between measures and calculated columns.

Calculated Column

A calculated column is calculated for each row in a table.

Measure

A measure calculates a value when it is used in a report and can change according to filters.

For example, an **Average Resale Price** measure can change when the user selects a particular product or brand.

For dashboard KPIs, measures are very useful.

---

9. Filters and Slicers

Filters help us show only the data we need.

For example:

Brand = Lenovo

will show information related to Lenovo.

Slicers allow the report user to select filters directly from the dashboard.

For EchoChain, possible slicers include:

- Product
- Brand
- Component
- Condition
- Year

This makes the dashboard interactive.

---

10. Power BI Visualizations

I learned that different visuals are useful for different types of analysis.

Card

Used for important numbers or KPIs.

Example:

Total Products: 250

Bar / Column Chart

Used to compare products or components.

Example:

Warranty Failures by Component

Line Chart

Used to show trends over time.

Table

Used to show detailed information.

Matrix

Used to show grouped information.

Scatter Chart

Can be used to study relationships such as:

Failure Rate vs Resale Value

---

11. Drill-down and Drill-through

Drill-down

Drill-down allows the user to move from a higher level to a lower level of information.

Example:

Product → Component → Individual Component

Drill-through

Drill-through allows the user to select a product and move to another page containing more detailed information.

For example, a product detail page can show:

- Warranty failures
- Resale price
- Components

- Refurbishment information
- Circularity Score

---

12. Conditional Formatting

Conditional formatting helps highlight important values.

For example:

- High warranty failures
- Low resale value
- High resale value
- High circularity

This makes it easier for users to identify important information quickly.

---

13. Dashboard Design

I learned some basic dashboard design practices:

- Keep important KPIs at the top.
- Use clear titles.
- Do not put too many charts on one page.
- Keep the design simple.
- Use consistent formatting.
- Leave enough space between visuals.
- Focus on the business question.

The main purpose of the dashboard is not just to make it look good. It should help the user understand the data and make decisions.

---

14. Power BI Service

I also learned the basic purpose of Power BI Service.

The general flow is:

Power BI Desktop → Publish → Power BI Service → Workspace → Report

Important concepts include:

- Publishing
- Workspaces
- Sharing
- Reports
- Dashboards
- Data refresh

I only need the basic understanding of Power BI Service for this project.

---

15. Scheduled Refresh

Scheduled refresh means updating the Power BI report regularly when new data is available.

For example:

New Data → Data Source → Power BI Refresh → Updated Report

This can be useful for EchoChain when new secondary-market data becomes available.

---

16. What I Learned for EchoChain

Through this project, I am learning how Power BI can be used to analyze:

- Product information
- Component information
- Warranty failures
- Secondary-market prices
- Product resale value
- Depreciation
- Circularity-related metrics

The main goal is to convert the available project data into an interactive report that can help understand product lifecycle and refurbishment opportunities.

---

17. My Contribution to EchoChain

My main responsibility is the **Analytics & Visualization layer**.

My work includes:

- Learning Power BI concepts.
- Connecting Power BI with Databricks SQL.
- Understanding Power Query.
- Learning data modeling and relationships.
- Learning DAX and Power BI measures.
- Planning EchoChain KPIs.
- Creating charts and KPI cards.
- Using filters and slicers.
- Learning drill-down and drill-through.
- Designing an executive-friendly dashboard.
- Understanding how to present business insights.

The data engineering work such as web scraping, PySpark processing and Delta Lake is handled as part of the team's other responsibilities. My focus is on the Power BI analytics side.

---

18. My Learning Flow

My Power BI learning for this project follows this order:

Power BI Basics

↓

Databricks Connection

↓

Power Query

↓

Data Modeling

↓

DAX

↓

Measures and KPIs

↓

Visualizations

↓

Slicers and Filters

↓

Dashboard

↓

Business Insights

---

19. Final Learning Summary

Through EchoChain, I am learning how a Data Analyst can use Power BI from connecting data to creating the final dashboard.